Clustering Hotel and Restaurant Reviews Using Sentiment Analysis

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Abstract
Platforms such as Booking.com and TripAdvisor site offers millions of valuable customer reviews on various hotels and restaurants. While the reviews are useful for both customers and service providers, the lack of structure makes it difficult to successfully harness the information. In this project, we solved the problem of automatically grouping hotel and restaurant reviews into meaningful sentiment-based clusters. This is important because clustering reviews can help customers to make quicker decisions. Also allow businesses to track service-related issues and strengths. We were able to solve this problem by employing clustering techniques like K-means, Hierarchical Clustering, and DBSCAN in addition to text preprocessing techniques like tokenization, stop word removal, lemmatization, and TF-IDF vectorization. The results were visualized, and dimensionality was decreased by using PCA. Our solution outperforms existing single-method approaches. It integrates multiple clustering techniques, capturing both broad sentiment groups and fine-grained hidden clusters.

Keywords*—sentiment analysis, text clustering, tf-idf, hotel reviews*

I. INTRODUCTION

Reviews by customers and clients about certain hotels and restaurants are stored on applications such as TripAdvisor and Booking.com which are online platforms. These reviews are stored on the platforms in huge amounts, but they are extremely difficult to categorize. to streamline the dividing and comparison process. In addition, business owners face similar issues where understanding the recurring themes from the feedback becomes a challenge. This project attempts to solve the issues by focusing on the sentiments expressed in the reviews and clustering the reviews on sentiments. The objective is to automate the process where the reviews are systematically divided and the extracted sentiments are classified. The system involves processes such as natural language preprocessing, TF-IDF feature extraction, and unsupervised grouping clustering algorithms.

II. LITERATURE REVIEW

Online hotel review mining has become fundamental in the comprehension of customer satisfaction and service quality improvement. Wang et al [1] explore big hotel review data as an attempt to fish out users' preferences and sentiments. According to them, online reviews carry a lot of valuable information on behavior and preferences that could be used for the betterment of services. Their approach begins with standard text-mining preprocessing (tokenization and feature extraction) then unsupervised clustering methods are applied. Several clustering algorithms are compared where it is found that K-means++ provides the best clustering performance, hence making possible coherent positive and negative categories of reviews which further divides evaluation information to make service personalization even more specific. Wang et al. will be followed because qualitative results are those that give an answer to whether the general delivery of the task has been achieved, but it should be noted that they did not present any quantitative evidence (for example: silhouette scores, cluster purity) and did not use external labeled data for cluster quality validation. Their method has, however, been recognized as one that runs fast and can easily be adapted for big data - where a detailed semantic relationship does not need to be comprehended.

In a related but broader work, probabilistic topic modeling was used by Guo et al. to analyze tourist satisfaction. They used 266,544 reviews by 25,670 hotels as input data for the Latent Dirichlet Allocation (LDA) model to elicit satisfaction's latent dimensions. The model comes up with thirty topics subsequently mapped into nineteen controllable service dimensions so varied as Staff Service, Bathroom Quality, and Dining Experience. Human evaluators validated these mapping exercises with Jaccard indices of 0.66 and 0.76-that is an impressively close match between LDA-derived topics and expert coding-this study also employs perceptual mapping plus regression analysis to see what effect which dimension has on overall satisfaction across hotel classes plus demographics contacted. It reveals gender-specific preferences (e.g. women mention homeliness and resort facilities more frequent) and rushes on the importance of service quality. Though LDA is interpretable and generalizable, that insight comes at a heavy computational cost-accompanied by the need for careful topic number selection and manual interpretation for incoherent topics.

Both studies utilize unsupervised text-mining, though intended for decidedly different analytical purposes. Wang et al. (2023) apply clustering of reviews which generates broad groupings based on the sentiments expressed, while Guo et al. (2023) allow a fine-grained thematic decomposition of review text enabled by their use of topic modeling. Clustering is presented as an easy and scalable approach to quickly segment reviews, whereas LDA provides semantically enriched insights at the cost of more computing resources. It would be strong if both approaches worked together: perhaps using clustering to do a high-level sentiment segmentation followed by topic modeling within clusters so that more fine-grained satisfaction factors can be revealed. Together, these works prove text mining through either clustering or topic modeling capable of deriving actionable intelligence from hotel review data in support of improving service quality and delivering personalized user experience.

III. METHODOLOGY

Data collection and loading: The dataset of user reviews (CSV) was read into R (read_csv). The columns were inspected, and the review text column was selected for processing. Basic checks were done for column names, row counts, and missing values. Empty reviews were removed or imputed to ensure a clean dataset.

Text cleaning and normalization: Each review was converted to lowercase. Contractions were expanded (replace_contraction). Emojis and emoticons were converted to text tokens (replace_emoji, replace_emoticon). Non-alphabetic characters were removed using regular expressions. Extra spaces were trimmed (str_squish). This produced clean plain text.

Tokenization and lexical normalization: Cleaned text was split into words (tokenize_words). Stopwords were removed (stopwords("en")). Words were normalized by stemming (wordStem) and lemmatization (lemmatize_words). **Spell correction and document assembly:** Misspelled words were checked and corrected (hunspell_check / hunspell_suggest). Tokens were then rejoined into cleaned text for vectorization.

Document-term representation and weighting: The cleaned text was stored as a corpus (VCorpus) and converted to a document–term matrix (DocumentTermMatrix). TF–IDF weighting (weightTfIdf) was applied. Sparse terms were removed (removeSparseTerms) to reduce noise.

Dimensionality reduction: The TF–IDF matrix was scaled, and principal component analysis (prcomp) was applied. The first 20 principal components were kept for clustering and visualization.

Clustering (three algorithms): Three clustering methods were used:

1. **K-Means** with k = 5 and a fixed seed.
2. **Hierarchical** clustering with Euclidean distance and Ward's method (hclust, method = "ward.D2"), cut into k clusters.
3. **DBSCAN** on scaled TF–IDF (dbscan), using chosen eps and minPts values. Cluster labels were added to the original dataset.

Visualization and inspection: Clusters were visualized by projecting onto the first two principal components with ggplot2. Scatter plots showed cluster membership, and dendrograms showed structure. Sample reviews were inspected to interpret cluster meaning.

Evaluation and interpretation: Cluster quality was checked by looking at cluster sizes, separation in PCA space, and DBSCAN noise points. Additional checks included silhouette scores and the elbow method. Clusters were interpreted as themes or sentiment groups, showing how hotels or restaurants could use insights for improvements.

Reproducibility and implementation notes: All steps were done in R using readr, tm, textclean, tokenizers, textstem, hunspell, SnowballC, slam, cluster, dbscan, and ggplot2. Random seeds were fixed for reproducibility. Parameters like sparsity threshold, number of PCs, k, eps, and minPts were tuned by inspection and can be refined further.

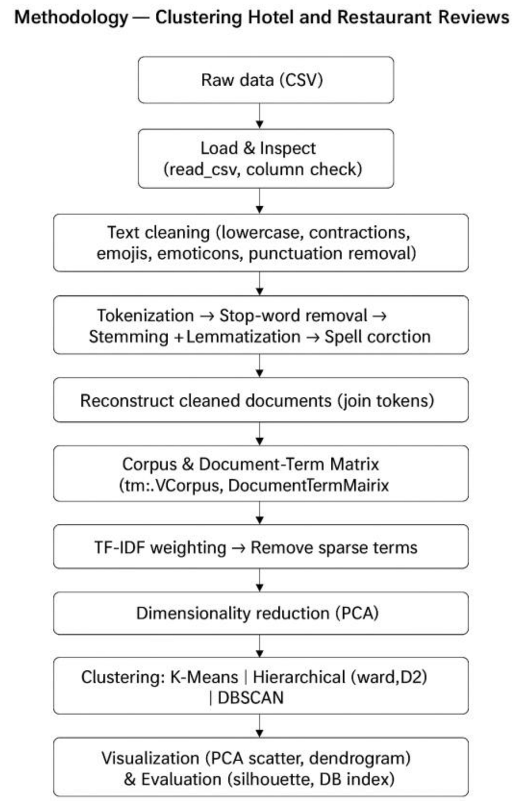

Figure 1: "Methodology Process"

IV. IMPLEMENTATION

The system was built in R and R studio. To accomplish specific objectives in distinct stages of the project, various libraries such as readr, stringr, tidytext, dplyr, SnowballC, and textstem as well as textclean, hunspell, tidyr, dbscan, ggplot2, and factoextra were installed and loaded. Each of these libraries served specific functions across the project such as data input/output and text cleansing. This project was represented by a file of cleaned CSV data from TripAdvisor containing customer reviews about restaurants and hotels, the reviews were modified to lowercase and afterwards cleaned of the text containing URLs, email addresses, punctuation, numbers, and extraneous spaces. When the reviews were cleared, the reviews were tokenized and stop words were removed according to the stop word list along with the automated system that performed lemmatization and reduced words to their base form with the lemmatize words function to increase the semantic consistency within the text. Full forms of contractions such as "can't" were converted, emojis were transformed to words and a basic spelling correction system was demonstrated using hunspell.

After preprocessing, the reviews were transformed into a document-term matrix using Term Frequency–Inverse Document Frequency (TF-IDF). The TF-IDF values were computed with the bind_tf_idf function, and the results were pivoted into a wide format using the pivot_wider function from tidyr to create a structured matrix suitable for clustering. The resulting TF-IDF matrix was then scaled to normalize the feature values.

Three algorithms were applied: K-means, Hierarchical Clustering, and DBSCAN for clustering. In K-means, the number of clusters was set to k = 5. And a random seed value of 123 to ensure reproducibility and 25 random starts (nstart = 25) to optimize stability of the results. Hierarchical Clustering was performed using Euclidean distance as the metric. Ward's method (ward.D2) for linkage, and the dendrogram was cut into five clusters to match the K-means setting. For DBSCAN, the minimum number of points was set to 5 (minPts = 5). The epsilon parameter was initially set to 1, based on the inspection of the k-nearest neighbor distance plot.

At last, reduction was performed using Principal Component Analysis (PCA) to project the high-dimensional TF-IDF features into two dimensions for visualization. The first two principal components were extracted and used to plot the results of K-means, Hierarchical Clustering, and DBSCAN in scatter plots. Each plot was created with ggplot2, showing how documents (reviews) were grouped and colored according to their assigned clusters. This visual representation made the results more interpretable and allowed for direct comparison of how each clustering algorithm segmented the dataset.

V. RESULT ANALYSIS

In the experimental results the clustering approaches can be clearly interpreted through the PCA projection diagrams. DBSCAN clustering was applied, the outcome showed that the algorithm was unable to form meaningful clusters from the dataset, and all points were grouped into a single cluster. This result suggests that DBSCAN could not successfully capture the structure of the data. Due to the chosen parameters such as epsilon and minimum points, the data distribution did not suit the density-based clustering assumption. For the lack of cluster differentiation here limits the usefulness of DBSCAN for this dataset.

For hierarchical clustering produced a more informative result by dividing the dataset into multiple clusters. As visualized in the PCA projection, the data points are spread across five visible clusters. Some overlapping regions are present. It indicates that hierarchical clustering was able to uncover subgroups within the data. But the separation between certain clusters was not entirely clear. Overlap may be assigned to similarities in the feature space or limitations of the hierarchical method in handling large, high-dimensional datasets.

K-means clustering provided the clearest and well-separated clusters among the three approaches. The PCA visualization shows five clusters that are more clearly partitioned compared to hierarchical clustering. The limits between groups are visibly sharper, suggesting that K-means successfully captured the underlying structure of the data. Some minor overlap still exists at the cluster borders, which is expected given the natural variability of the data and the projection into two dimensions. The clustering performance of K-means appears superior in terms of compactness

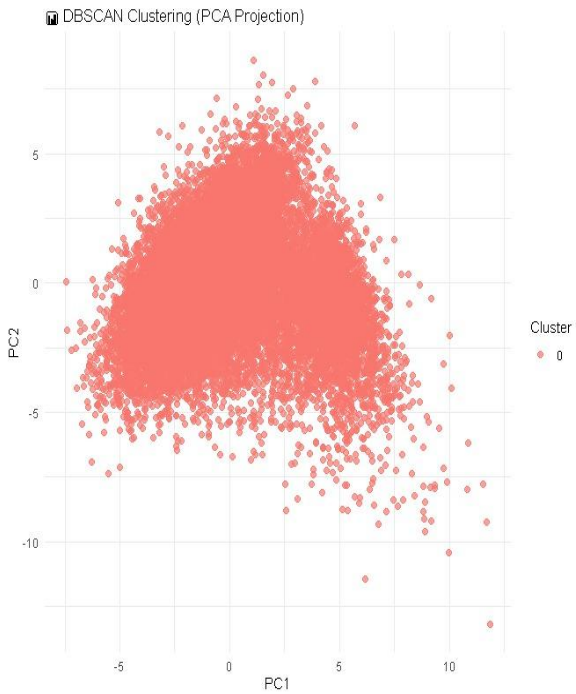

Figure 2: "DBSCAN Cluster Distribution"

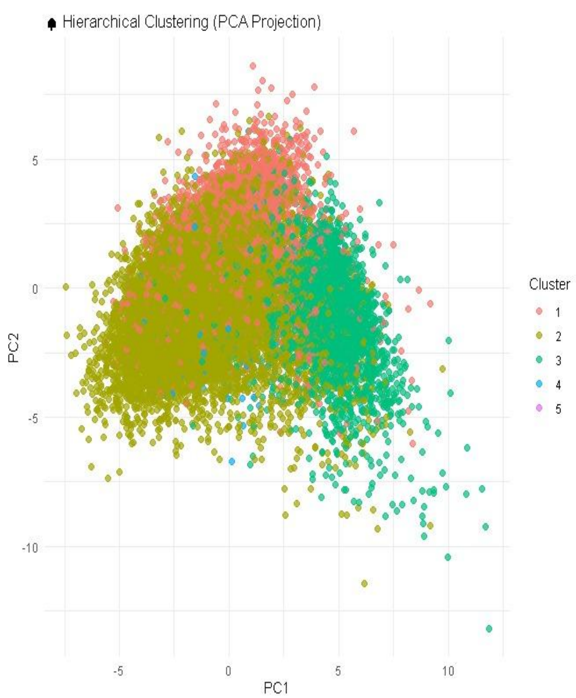

Figure 3: "Hierarchical Cluster Distribution"

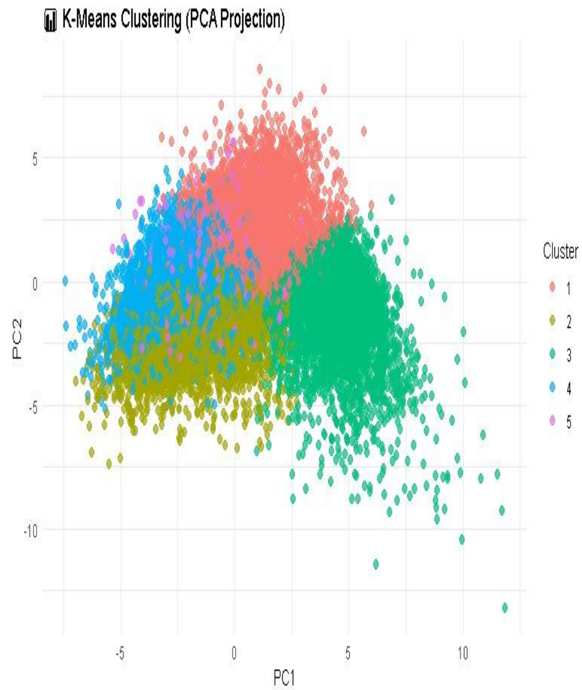

Figure 4: "K-means Cluster Distribution"

VI. CONCLUSION

Clustering techniques were applied to analyze and uncover meaningful patterns within the dataset. The results were visualized through PCA projections for better interpretability. Three popular algorithms—DBSCAN, hierarchical clustering, and K-means clustering—were compared to evaluating their ability to separate the data into coherent groups. From the experiment DBSCAN was unable to identify distinct clusters and grouped all points into a single cluster. Limitation under the given parameter settings and data distribution. Hierarchical clustering provided an improvement by dividing the dataset into multiple clusters. Some overlaps reduced the clarity of separation. K-means clustering carried the most promising outcome by forming well-separated and compact clusters. It makes the most effective method for this dataset. This study has shown the importance of carefully selecting clustering algorithms based on data characteristics and application requirements. By comparing these methods, the work provided valuable insights into the strengths and weaknesses of different clustering techniques, which can guide future research and practical applications. The findings highlight that clustering is not only a powerful unsupervised learning tool for data exploration but also a key step toward meaningful knowledge discovery.

VII. REFERENCES

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